An AI-Based Financial Advisory Model Framework for Digital Investments among Mid-Income Filipinos

By

Julienne Dana T. Felipe
Michael Mario P. Kambayashi
Dylan Hope A. Salazar

Bachelor of Science in Computer Science
Mapua University - Makati, 2026-2027

A Thesis paper Submitted to the School of Undergraduate Studies in Partial Fulfilment of the Requirements for the Degree

Bachelor of Science in Computer Science

School of Information Technology
October 2026

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Chapter 1

INTRODUCTION

Background of the Study

Financial Technologies are tools that provide consumers automated financial services to support business operations in the digital world. In partnership with the financial sector, tech companies continue to revolutionize and deploy software that assists in investments and transactions in both micro and macro levels. Positive impact was measured as some digital platforms and mobile services have proven their accessibility among Philippine marginalized communities [12]. However, even with these advancements, a large portion of the Filipino population continues to face financial illiteracy as business advising and support remains limited to higher societal classes [13].

To be financially literate, an individual must have the ability to make good financial decisions in budgeting, saving, investing, and applying for insurance. Unfortunately, a survey found that only 25% of Filipino adults were found to be financially literate as of 2014, placing the country among the lowest globally [10]. This percentage implies that many households struggle in long-term financial planning, management, and investment. With the absence of affordable financial education, gaps develop between an individual's capacity and accessibility towards financial services. These gaps include inadequate infrastructure, digital and financial literacy, and limited access which primarily affected underprivileged communities [11]. Although several news reports suggest that factors that contribute to financial literacy have improved in the Philippines throughout the years, disparities remain.

With the emergence of Artificial Intelligence (AI) into financial services, current

available models have demonstrated scalability and personalization for financial advising. Financial Technologies use Artificial Intelligence models, frameworks, and algorithms as the foundation for robo-advisors, chatbots, and predictive analytics in providing services for users in their financial decisions [14]. In the local context, Velez [2025] emphasized that future work and policies must integrate multidimensional strategies to achieve equitable financial inclusion in the Philippines [14] and ultimately contribute in improving the population's financial literacy rate. However, despite the tools made available by AI-driven financial technologies, the lack of trust towards automated financial advice still persists. This suggests the need for a localized, interpretable, and user-centered advisory system.

Gap/Opportunity

Takayanagi et al. [2025] examined whether Artificial Intelligence, specifically Large Language Models (LLMs), can support financial advising using their generative capabilities [4]. Although Zhang et al. [2025] noted that current models tend to generate biased recommendations despite debiasing techniques [9], Takayanagi et al. [2025] argued that these tools can effectively personalize recommendations and acknowledge investor preferences [4]. Furthermore, LLMs rely heavily on user data to generate financial insights that can influence an investor's decision-making. With this evidence, several researchers attempted to design their financial advisory models using different methodologies.

Beyond LLMs, researchers have explored various AI architectures for financial advising. For instance, Asemi et al. [2024] designed an investment-type recommender

system by integrating an Adaptive Neuro-Fuzzy Inference System (ANFIS) with a Multimodal Neural Network (MNN) to process investor demographic, financial, and personality profiles [1]. Similarly, some researchers developed financial recommendation algorithms that propose adaptability [7][8], forecasting [6][26], and planning [5][6] for financial investing. However, these models have their limitations due to their methodological constraints. Some studies have yet been validated reliably despite producing substantial results [5][6] while others used curated, small, old, or zero datasets [7][8][26]. These research designs limit their models from addressing investment feasibility [6], long-term validation [5], low-resource and localization [1][7][8] opportunities.

Asemi et al. [2024] defined 8 categories that influence the recommendations that these models produce. These categories can be classified into 3 major factors: demographic, financial, and behavioral information. Considering that the Philippines was among the countries with a low financial literacy rate in 2014 [10]—although it has been noted that the country may have improved significantly—continuous financial education towards micro-investments must be provided, especially for marginalized Filipino communities.

Currently, the development of localized, interpretable, hybrid NLP-forecasting models for Filipino micro-investing remains notably underexplored in existing literature. With this gap in mind, this study proposes a localized advisory model framework that generates personalized micro-investment recommendations by integrating behavioral assessment, market forecasting, and interpretable decision-making tailored for Filipino users.

Statement of the Problem

This study proposes the development of a hybrid AI-based financial advisory framework that integrates financial-capacity clustering, natural language processing, market forecasting, and interpretable rule-based decision-making. Specifically, the study aims to answer the following questions:

1) What interpretable mechanisms allow the model framework to generate interpretable and data-driven financial guidance with limited Filipino financial datasets?
2) How can the model generate personalized micro-investment recommendations while considering demographic, financial, and behavioral factors?
3) How can the hybrid model's accuracy, interpretability, and transparency performance be evaluated to measure its effectiveness?

Research Objectives

The main objective of the study is to develop a localized, integrated model framework for financial advising and recommendation on micro-investments. The framework shall be designed to achieve the following:

1) To develop a hybrid advisory framework that integrates language understanding, market forecasting, and interpretable recommendation generation.
2) To integrate Fuzzy C-Means financial-capacity clustering, XLM-RoBERTa behavioral risk classification, Prophet market forecasting,

and an interpretable rule-based feasibility assessment to generate personalized micro-investment recommendations.

3) To evaluate the framework using model-specific quantitative metrics (RMSE, MAE, MAPE, FPC, Silhouette Score, accuracy, precision–recall, F1-score) and a qualitative user evaluation of recommendation clarity, transparency, relevance, and perceived usefulness.

Significance of the Study

This study shall primarily contribute to the world of financial technology, specifically catered towards Philippine financial companies. Research suggests that these advanced digital tools enhance customer experience, supporting fintech companies in the public expansion for a broader audience [15]. In comparison to other research, the model's localization feature will benefit fintech companies to support their business ventures in offering accessible, financial services that enhance their relationship between their customers.

Additionally, the study shall benefit middle-income individuals by exploring the use of an AI-based decision-support framework for micro-investment planning. The financial-capacity component evaluates the respondent's income, expenses, savings, and number of dependents, while the behavioral component identifies investment risk tolerance from open-ended survey responses. The market-forecasting component provides information regarding the projected direction and uncertainty of the Philippine Stock Exchange Index (PSEi). These factors are integrated through an interpretable rule-based feasibility assessment to produce personalized investment guidance. Rather

than executing financial transactions or guaranteeing investment returns, the framework is intended to support more informed and transparent financial decision-making. By facilitating this accessible, data-driven approach to personal finance, the system can assist in establishing sustainable financial behaviors [16] and ultimately contribute to decreasing the financial literacy gap in the Philippines.

The academia and research field is also a benefactor of this study. It shall extend on the growing literature towards Artificial Intelligence models while considering the ethical standards and responsible usage in using these tools for the advancement of financial technologies. By adopting a hybrid NLP-forecasting framework, the model shall further explore and validate different advanced machine learning and generative tools useful for financial advising and recommendation.

Scope and Limitations of the Study

The study population is limited to users from mid-income individuals residing in the Philippines. With that in mind, limited access to large-scale financial datasets may restrict the system's accuracy and generalizability. Moreover, the study is expected to span 6 months. The system will not handle actual money transactions but will instead provide financial guidance and recommendations. It also means that the AI framework will be developed and tested in a controlled, model prototype-level environment, and not deployed as a full production system. Limited computational resources can constrain the development of complex models and the exploration of advanced deep learning architectures. The study will also provide micro-investment decision-support guidance for middle-income individuals. Furthermore, the AI-generated recommendations serve only

as research-oriented micro-investment decision-support guidance and should not be interpreted as professional financial advice. Practical investment returns, financial outcomes, and other user-specific concerns, and real-time market fluctuations are beyond the scope of this study as these variables are highly dependent on the external market. The behavioral risk profile is inferred from respondents' open-ended survey responses and represents a research-oriented classification rather than a professionally administered psychometric risk assessment. Finally, while the NLP module leverages multilingual models, initial data gathering revealed that respondent answers were often highly truncated (e.g., 2-3 words) and predominantly English. While the system complies with the Philippine Data Privacy Act, ethical considerations such as long term accountability and algorithmic fairness, falls outside the scope of the study.

Chapter 2

REVIEW OF RELATED LITERATURE

Micro-Investing, Digital Platforms, and Financial Services

Micro-investing, according to Gonzalez-Arcos et al. [2023], is the practice of investing small amounts of money through financial instruments to acquire returns for additional income from the financial market. It encourages long-term financial participation and engagement [27] that trains individuals to develop good financial skills and awareness towards investment opportunities. With the help of online digital platforms, micro-investing has become more accessible to the general public, positively influencing households in their financial decisions and behaviors [28].

There are many factors associated with micro-investing. In a Computer Science related study, Asemi et al. [1] classified these into three major categories before designing their financial advisory system. The first group covers data on demographic information. A study from 2006 found that demographics significantly contribute to the financial viability of households in Asian countries. Particularly in the Philippines, they revealed that the disproportionate financial strain from larger household sizes limit an individual's capacity to pursue investment opportunities [25]. This implies that there's a direct relation between demographic variables such as number of dependents and financial viability. Similarly, household financial and behavioral profiles were also major elements. Desello and Agner [2023] and Mendez-Suarez et al. [2019] identified these financial and behavioral variables (such as income, savings, debts, educational attainment, recurring expenses; budgeting habits, saving consistency, timely payment of bills, investment experience, financial goal setting, impulsive spending and reliance on debt) as key determinants for financial inclusion

and decision-making [10][15][26].

Given the dynamic of these interacting factors, a simple manual assessment may be insufficient to determine micro-investment feasibility. However, with the recent innovations in financial technologies, opportunities in developing advanced financial approaches and services for micro-investing increased abundantly.

In Southeast Asia, online digital platforms that offer micro-investing opportunities have become trends that cater to emerging market investors. In the Philippines, financial technologies showed promising results as financial services continue to extend. Despite ranking among the countries with the lowest financial literacy rate in 2014 [10], some reports showed that these technologies accelerated during the COVID-19 pandemic due to the demand for cashless digital transactions [22]. For example, GCash made digital financial services available for Filipinos of all social classes. Such services include the facilitation of government subsidies (G2P), micro-savings (G-invest), insurance (G-insurance). micro-lending (G-Gives). By pursuing partnership programs with the Philippine government, financial inclusion, especially towards underserved communities [23], have significantly improved. However, local and global trends noted the indirect connection between micro-investment platforms and economic sustainability, suggesting that the impact of these digital technologies is still dependent on user education and institutional trust [24].

Despite improvements in mobile wallet penetration and digital infrastructure, institutional barriers hinder individuals from adopting advanced technologies and tools. These gaps suggest the need for a computational model capable of integrating and analyzing financial feasibility profiles simultaneously.

AI-Powered Financial Advisory Models and Related Systems

Research on AI-driven financial advisory systems has examined the development of automated financial services. For instance, a study revealed that robo-advisers offer a cheaper alternative for investment advising and recommendation compared to traditional practices. Although innovative, these basic AI tools have their limitations on the services they can only provide for customers. Moreover, the rise of AI has also raised ethical concerns generally on data privacy, algorithmic transparency, and standard management of personal financial information [17]. As AI grows in the world of financial technology, so should the design of existing or future models evolve based on these limitations.

In the context of micro-investing, several studies have attempted to design frameworks that can recommend investment opportunities and provide financial advising. For example, Asemi et al. [2024] designed a hybrid model using an Adaptive Neuro-Fuzzy Inference System (ANFIS) and a Multimodal Neural Network (MNN) to give personalized investment-type recommendations. Although their system performed well in predictions and classifications, they only covered a small population of (demographic, financial, and personality) investor profiles which may not be applicable to different or larger populations. A similar study also developed a financial recommendation algorithm by using big data analysis and user interest pattern with Fuzzy K-Means algorithm [7]. The algorithm performed outstandingly in clustering algorithms, producing a lower MSE and a higher precision and recall. However, Yang et al. [2024] observed its complexity despite using zero datasets and claimed that their proposal may not account to market conditions [7]. Financial automated advising powered with Artificial Intelligence was also tested for the copper market [26]. In Méndez-Suárez et al.'s [2019] study, they used copper prices to train an

Artificial Neural Network model to forecast trends 5 days in the future. Although criticized by using an old dataset (daily copper prices from London's 2006-2010 economic period), their framework produced accurate price forecasts with an average of 29.1% financial return over 24 days and captured market inefficiency during the time period [26].

There were also similar architectures that applied different approaches towards financial systems. For example, the proposed WealthGuardian by Walia et al. [2025] was trained to predict future expenses and recommend insurances to a user. Using integrated financial data from bank API's with Support Vector Machines and Random Forest, the proposed system performed its intended purpose of delivering real-time expense forecasts with personalized advice. However, performance evaluations have yet been tested and may further struggle with complex patterns to assess practical accuracy and impact [6]. Another approach was also implemented by Lee et al. [2024] towards the stock market. They used dynamic user preferences and stock market behaviors as dimensions to design their temporal graph network to personalize stock recommendations and diversify portfolio investments. Test results showed an additional 3.45% performance, surpassing established baseline models on balancing recommendation relevance and investment profitability. The study does not explicitly address the transparency or interpretability of its decision-making process, which may hinder user trust and limit practical adoption in real-world financial settings [2]. On the other hand, Manasa et al. [2025] developed a financial management system for personal use. Using linear regression, the system was able to generate predictive insights and timely alerts that forecast user expenses for budget control and expense monitoring. Although it produced its intended results, relying primarily on this regression tool hasn't effectively captured the complex financial patterns that limit the accuracy of expense predictions [3].

Natural Language Processing, Forecasting, and Generative Advising

Natural Language Processing (NLP) enables AI systems such as chatbots, sentiment analyzers, and recommender algorithms to interpret financial and trading texts. In the financial context, NLP models are used to extract tone, detect market signals, and process large volumes of unstructured data such as news articles, earning calls, and investment reports. For instance, Mino et al. [2025] examined the application of transformer-based models like BERT in financial tone analysis and found that FinBERT outperformed traditional machine learning approaches by leveraging vocabulary and contextual patterns unique to financial language [20]. The results showed that FinBERT demonstrated an improvement on classification accuracy across multiple financial benchmarks. However, despite its strong performance, the study focused on high-resource languages which limits the applicability to multilingual or low-resource language environments.

Large language model evaluations highlight the gap in performance for low-resource languages compared to English and other widely studied languages [21]. Montalan et al. [2025] examined the linguistic challenges of applying NLP tools to Tagalog due to the scarcity of native linguistic resources that hinder the development of accurate Filipino language models. They claimed that these tools often rely on translation pipelines or multilingual models as substitutes for native language data [21]. While existing studies identify the structural challenges, most research focuses on general language understanding rather than specialized areas, leaving localized financial NLP systems underexplored.

Moreover, AI systems also integrate predictive modeling to transform extracted data into practical financial guidance. Talasila [2024] proposed an AI-driven personal finance management system that combines ARIMA and LSTM models that integrates machine

learning algorithms for predictive budgeting, categorizing transactions, and setting saving goals. The system, with data from 1,000 users in a span of 6 months, was reported to have produced a significant positive impact on user financial capacity and behavior: 22% increase in average monthly savings, 45% reduction in reported financial stress, and a 92% success rate in avoiding late bill payment fees [5]. These percentages imply that forecasting models can operationalize financial data into practical decision support tools. Similarly, Walia et al. [2025] emphasized the integration of NLP and automated generative advising. The approach involves decision trees and rule-based logic to generate personalized financial plan recommendations tailored to user profiles and spending behavior [6]. They acknowledged their model's relevance in generating recommendations compared to generic financial suggestions. However, despite these promising directions, the model's impact has yet been proven and measured to be practically accurate and positive.

These developments in personalized advising have since been expanded through the application of large language models. Yang et al. [2023] introduced InvestLM, a language model fine-tuned on LLaMA-65B using curated financial datasets. Their model consists of financial data from exam questions about finance, SEC filing and finance forum queries that supported its high performance better than financial NLP benchmarks and other existing models (specifically GPT-3.5/GPT-4) on investment related queries. However, despite these strong results, the performance primarily reflects improvements in questions-answering accuracy rather than portfolio performance or risks [8]. In another approach, Zhi et al. [2025] conducted an analysis of approximately 567,000 recommendation outputs spanning stocks, mutual funds, cryptocurrencies, and savings products. Using a prompt-and-analysis pipeline to measure product frequency distributions across generated outputs, the study found that

large language models resulted in product bias, appearing to over-recommend certain financial instruments despite debiasing techniques [9]. These findings suggest that generative financial models may unintentionally influence asset allocation patterns in ways that could raise fairness and market stability concerns. However, despite the comparative study's scope, the study only used synthetic data which may not adequately account for the impact of biased recommendations on actual investor behavior.

Predictive analytics and forecasting models assist automated investment advice systems by predicting market trends and providing personalized investment recommendations. Facebook's Prophet, a time series model, addresses seasonality and variation in trend using an additive model. Empirical findings show that Prophet outperforms traditional ARIMA models in forecasting stock prices, producing lower forecast errors in banking sector data [19]. Such models can further inform automated advisors by providing more accurate projections of asset returns and risk metrics.

Limitations of Personalization, Bias, and Trust in AI

While AI advisers seem promising, scholars have persistently cautioned against their pitfalls. Such prominent challenges that these advisers struggle with include data privacy, transparency, and algorithmic bias. Surveys suggest that a significant number of consumers do not feel safe sharing personal information with an algorithm with the potential of being hacked or having their data misused [18].

Transparency is also essential. A recent study found that lack of transparency was the most common reason Filipinos and others are hesitant to use robo-advisors [18]. Even if the provider discloses the technicalities of its AI, a typical user will rarely have background

knowledge to interpret this information meaningfully. Without transparency, the users would be left unaware why an investment was specifically recommended. The lack of clarity can also hide system flaws and conflicts of interest.

Bias and fairness present other issues. For instance, financial products usually involve significant stakes that require long-term commitments. When these models are used, they tend to be sensitive to biased recommendations due to user profile data [28]. Recommender systems in finance must contend with sparse data and are subject to biases based on popularity. Recent research on LLM-generated investment advice found that the algorithmic "product bias" has self-reinforced favoritism for certain stocks or funds (for example, tech giants), which the AI intelligence applied without context. Risk profiling tools can also reproduce bias based on gender or culture without proper modeling controls during the design process as shown by recent literature [25]. The literature suggests incorporating a range of bias mitigation strategies (such as fairness-aware algorithms and diverse training data) and developing methods to carry out ongoing assessments. Hybrid recommendation methods and explainable AI techniques are being explored to ensure personalization does not inadvertently exclude or mislead users.

Chapter 3

METHODOLOGY

This section covers the methods and procedures in developing the AI-based financial advisory system framework for digital micro-investments among mid-income Filipinos. It outlines the data gathering and preparation process, designing the hybrid NLP-forecasting, soft clustering and performance validation.

The implementation and experimentation of this framework shall utilize Python within the Jupyter Notebook environment.

Conceptual Framework

The study proposes a framework that integrates established modeling approaches to its core components to create a hybrid AI-based financial advisory system. The framework works through three parallel data processing pipelines that converge to generate investment feasibility assessments.

The proposed framework consists of three analytical branches. First, the Financial Capacity Module applies Fuzzy C-Means (FCM) clustering to structured demographic and financial variables to derive relative Low, Medium, and High Financial Capacity profiles. Second, the Behavioral Analysis Module applies an XLM-RoBERTa zero-shot classification model to open-ended user responses directly related to investment risk preference and reaction to financial loss, producing Low Risk Tolerance, Neutral, or High Risk Tolerance profiles. Third, historical market data from Philippine financial sources undergo forecasting using the Prophet time-series model after basic cleaning [29]. The resulting 30-business-day forecast and uncertainty intervals are then converted into Positive, Neutral, Negative, or

Volatile market signals.

The outputs of the three modules are then combined using a complete rule-based feasibility matrix. Each combination of behavioral profile, financial capacity, and market signal maps to a predefined rule adjustment. A neutral baseline feasibility score of 0.50 is modified by the matched rule, producing a feasibility score and an interpretable rule result. The final recommendation module translates this result into user-facing investment guidance without performing a second independent decision process. Furthermore, a summary of the model is outlined in Figure 3.1.

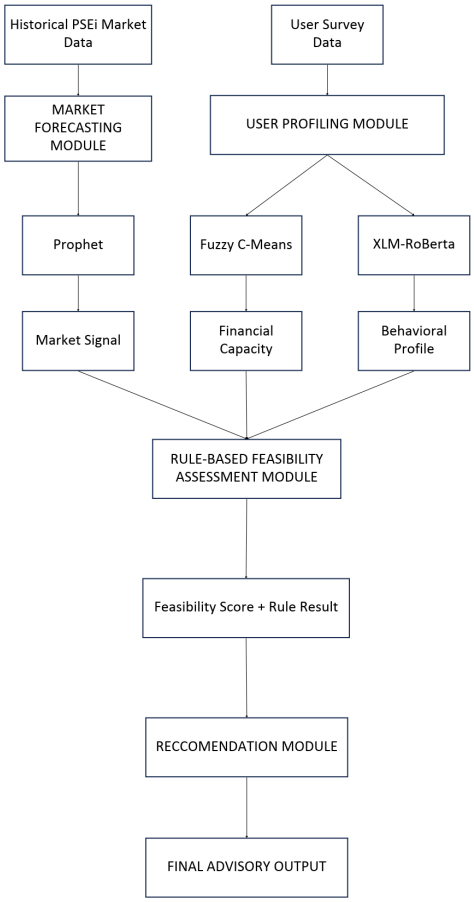


Figure 3.1. Conceptual Framework Pipeline

Data Collection

The study employs two primary data sources: user survey data and historical Philippine Stock Exchange Index (PSEi) market data, which provide the information required by the framework's three analytical branches. This dual approach enables the AI-Based Financial Advisory System to analyze not only the quantitative dynamics of the Philippine financial market, but also the contextual and behavioral factors that influence individual investment decisions. By combining these market and user dimensions, the system is designed to produce data-driven, context-aware investment recommendations that are interpretable, localized, and aligned with real-world economic conditions.

The first component, Part A - Historical PSEi closing-price data are obtained from Investing.com and manually provided to the experimental prototype as a CSV file. Manual file input is used to preserve reproducibility by allowing the same historical dataset to be reused across experimental runs. The market dataset contains date and closing-price observations. When the source labels closing prices as Price, the variable is standardized to Close before processing. The date and closing-price variables are subsequently transformed into the ds and y structure required by Prophet. The collection period covers January 2021 to August 2026, ensuring that both mid-pandemic and post-pandemic market behaviors are captured to reflect realistic fluctuations in the Philippine financial landscape.

The second component, Part B - User data are collected through an online survey using purposive sampling among Filipino respondents within the study's target monthly gross-income range of ₱24,060 to ₱144,360. This demographic represents the mid-income bracket of the Filipino population who are generally capable of saving or investing but often

lack access to affordable and personalized financial advisory services. The survey contains demographic, financial, behavioral, and contextual questions. Financial variables include average monthly gross income, monthly expenses, net savings, and number of dependents. Open-ended behavioral questions examine respondents' attitudes toward investment risk, preferred financial choices, investment goals, and reactions to potential investment losses. The survey gathered a range of attributes divided into three major categories:

1) Demographic data - including age, sex, civil status, employment status, educational attainment, and number of dependents;
2) Financial data - such as monthly gross income, monthly expenses, net savings, educational expenses, and current loans or debts; and
3) Behavioral data - covering saving habits, investment-seeking behavior, spending behavior, financial decision-making, investment goals, risk preference, and reaction to a hypothetical investment loss.

Although several behavioral questions are retained for descriptive and contextual analysis, the final behavioral risk-classification module specifically uses Question 7, which asks about preference between traditional banking and higher-risk investments, and Question 9, which asks how the respondent would react to a hypothetical investment loss. These two questions were selected because they directly represent investment risk preference and loss tolerance. Missing data are handled according to the semantic meaning of each variable and the requirements of each analytical module rather than by removing every respondent with any unanswered survey item. Blank responses for Number of Dependents are interpreted as zero based on the survey context. In contrast, respondents lacking the core financial variables required by Fuzzy C-Means are excluded from the FCM branch. Missing or non-informative

Q7 (Question 7) and Q9 (Question 9) responses are handled separately by the behavioral module. The behavioral portion of the survey also included open-ended responses to capture authentic user expressions of goals and concerns. (e.g., “prefers traditional banking, savings accounts or other low-risk financial options”). These textual responses were processed using XLM-RoBERTa, an NLP model designed by Meta (FaceBook) trained on 100 different languages, to interpret Filipino, English and Taglish language inputs and extract semantic features related to investment risk preference and loss tolerance. Data gathering occurred between May to September 2026, preceded by a pilot test to assess the clarity, reliability, and usability of the questionnaire.

This user dataset plays a critical role in building the behavioral intelligence of the system. Using an XLM-RoBERTa zero-shot classification pipeline, the system processes natural language statements to capture contextual meaning, risk perception, and intent, directly categorizing users into distinct behavioral profiles (Low Risk Tolerance, Neutral, or High Risk Tolerance). These behavioral profiles are then evaluated alongside the user's dominant Financial Capacity category and the market forecasting outputs from Prophet. By integrating these three independent dimensions through a rule-based feasibility matrix, the hybrid model generates personalized investment feasibility scores that align with both user circumstances and market realities.

Data Preprocessing

Data preprocessing is divided into two independent workflows based on data type: (1) user profile data and (2) historical market data. Survey preprocessing prepares structured financial variables for Fuzzy C-Means and open-ended behavioral responses

for XLM-RoBERTa, while market preprocessing converts the historical PSEi closing-price series into the time-series structure required by Prophet.

Market Time-Series Preparation

The historical PSEi dataset is prepared according to Prophet's required time-series format. The date variable is converted to datetime format and renamed ds, while the closing-price variable is converted to numeric form and renamed y. Observations are sorted chronologically. If isolated closing-price values are missing, forward filling is applied using the most recent available price. No additional demographic or financial variables are included in the Prophet model because the forecasting branch operates independently on the historical PSEi closing-price series.

Financial Capacity Preprocessing and Fuzzy C-Means Clustering

To prepare the data for Fuzzy C-means (FCM) clustering, the Financial Capacity Module applies the algorithm to four core variables: Number of Dependents, Average Monthly Gross Income, Average Monthly Expenses, and Average Monthly Net Savings. Educational expenses and debt values are excluded from the final clustering model due to a high volume of missing responses, which would mathematically destabilize the algorithm. Additionally, Age is excluded from the clustering process, as it is retained strictly for descriptive survey purposes rather than serving as a direct measure of financial capacity. During preprocessing, blank 'Number of Dependents' values are interpreted as zero based on the survey context, while respondents missing any of the core monetary variables are excluded entirely to preserve data integrity. To reduce the influence of extreme monetary

observations, Income, Expenses, and Savings undergo Winsorization (1st and 99th percentiles) followed by a Logarithmic Transformation (log1p). Finally, all four variables are rescaled using a Standard Scaler (Z-Score Normalization) so that differences in numerical scale do not dominate Euclidean distance calculations.

FCM is implemented using three clusters (C = 3) with a fuzziness coefficient of m = 1.5, an error tolerance of 0.005, a maximum of 1,000 iterations, and a fixed random seed of 42. FCM was utilized to circumvent traditional systems, which often use strict, hard-coded rules (e.g., "Income < ₱30,000 = No capacity to invest"). Such classifications fail to capture financial nuance. By generating fuzzy membership degrees, the algorithm indicates the degree to which a respondent mathematically resembles each cluster rather than forcing a single classification. The resulting fuzzy membership degrees are retained for interpretability and reporting, while the cluster with the respondent's highest membership degree is passed to the rule-based feasibility module as the dominant category. The numerical cluster identifiers produced by FCM have no inherent meaning, the clusters are interpreted after clustering using their original financial characteristics. A Capacity Interpretation Score is calculated using standardized cluster-level means:

$$\textit{Capacity Interpretation Score} = z(\textit{Income}) + z(\textit{Savings}) - z(\textit{Expenses}) - z(\textit{Dependents})$$

Higher income and savings naturally increase the score, while greater expenses and more dependents represent a financial burden and reduce the score. Because the system dataset is already strictly filtered to the target demographic of mid-income earners (₱24,060–₱144,360), these semantic labels represent relative financial capacity rather than

raw income brackets: [30];

1. **Low Capacity:** the lowest-scoring cluster, consisting of individuals experiencing the highest relative financial strain. This profile is characterized by the lowest average gross income combined with the highest number of dependents and high living expenses.
2. **Medium Capacity:** the intermediate-scoring cluster, consisting of high-earners who carry significant financial responsibilities. This group has the highest absolute average gross income, they also carry the highest monthly living expenses and a high number of dependents.
3. **High Capacity:** the highest-scoring cluster, consisting of individuals with strong relative financial capacity. This group is characterized by the lowest monthly expenses, the fewest dependents and a high proportional savings rate.

Behavioral Text Preparation

Open-ended survey responses are subjected to minimal text preprocessing prior to behavioral analysis. Text is converted to lowercase and excess whitespace is removed. Aggressive linguistic normalization, such as stopword removal, stemming, translation, or punctuation stripping, is not performed because the pretrained multilingual XLM-RoBERTa model is intended to interpret contextual language directly. The behavioral classification focuses specifically on Question 7 and Question 9. Responses containing no usable behavioral information, including blanks and placeholder entries such as “N/A” or “-”, are treated as missing rather than being forced into a behavioral category.

Behavioral Risk Classification

To extract a user's investment risk tolerance from unstructured, open-ended survey responses, the Behavioral Analysis Module employs a zero-shot classification pipeline utilizing the pre-trained joeddav/xlm-roberta-large-xnli model. Zero-shot classification allows the framework to assign behavioral categories by evaluating the semantic similarity between the user's text and descriptive labels, entirely bypassing the need to train a new supervised classifier on a limited survey dataset.

The module specifically evaluates two key survey items. Question 7 evaluates the respondent's explicit preference between safer financial options and higher-risk investments. Candidate behavioral meanings are mapped to three categories:

1. **Low Risk Tolerance**: preference for traditional banking, savings accounts, or other low-risk financial options.
2. **Neutral:** preference for diversified or moderate-risk investments such as mutual funds, or a combination of savings and investments.
3. **High Risk Tolerance:** preference for speculative or high-risk investments such as individual stocks or cryptocurrency for potentially higher returns.

Question 9 separately evaluates the respondent's behavioral reaction to a substantial hypothetical investment loss:

1. **Low Risk Tolerance**: selling the investment or avoiding further investment.
2. **Neutral:** holding the investment, observing the market, or waiting before making a decision.
3. **High Risk Tolerance:** buying more or continuing to accept substantial investment risk.

XLM-RoBERTa classifies Question 7 and Question 9 independently by generating a numerical zero-shot classification score for each of the three candidate behavioral profiles: Low Risk Tolerance, Neutral, and High Risk Tolerance. For a given response, the profile associated with the highest score is selected as the item-level classification. The highest score is retained as a diagnostic confidence value, while all three class scores are preserved for quantitative analysis. Thus, the open-ended behavioral response is first transformed into numerical model outputs before being represented categorically.

For a behavioral response x, let $S_L(x)$, $S_N(x)$, and $S_H(x)$ denote the XLM-RoBERTa scores for Low Risk Tolerance, Neutral, and High Risk Tolerance, respectively. The item-level behavioral classification is defined as:

$$C(x) = \arg\max \{S_L(x), S_N(x), S_H(x)\}$$

and its diagnostic confidence is:

$$Conf(x) = \max \{S_L(x), S_N(x), S_H(x)\}$$

The numerical scores are retained for reporting and model inspection but are not averaged to determine the final Behavioral Profile. Instead, the independently classified Q7 and Q9 categories are combined using the transparent Behavioral Profile Combination and Conflict Resolution Matrix shown in Table 3.1. When Q7 and Q9 agree, the agreed category is retained. When one response is Neutral and the other provides a directional Low or High classification, the directional response determines the final category. Direct Low–High conflicts are represented as Neutral. If only one valid response is available, that response is used; if neither is usable, no Behavioral Profile is assigned.

Table 3.1. Behavioral Profile Combination and Conflict Resolution Matrix

Q7 Profile (Investment Preference)	Q9 Profile (Loss Reaction)	Final Behavioral Profile	Decision Logic (System Reasoning)
Low Risk	Low Risk	**Low Risk**	Both responses indicate lower willingness to accept investment risk.
Neutral	Neutral	**Neutral**	Both responses indicate moderate or balanced risk behavior.
High Risk	High Risk	**High Risk**	Both responses indicate greater willingness to accept investment risk.
Neutral	**Low Risk**	Low Risk	The directional Low Risk classification is retained over Neutral.
Neutral	**High Risk**	High Risk	The directional High Risk classification is retained over Neutral.
Low Risk	Neutral	Low Risk	The directional Low Risk classification is retained over Neutral.
High Risk	Neutral	High Risk	The directional High Risk classification is retained over Neutral.
Low Risk	High Risk	**Neutral**	Contradictory directional classifications are represented as Neutral.
High Risk	Low Risk	**Neutral**	Contradictory directional classifications are represented as Neutral.
(Missing / Blank)	**Any Valid Profile**	(Uses Q9 Profile)	System relies on the single available valid response
Any Valid Profile	(Missing / Blank)	(Uses Q7 Profile)	System relies on the single available valid response
(Missing / Blank)	(Missing / Blank)	**(Excluded)**	No behavioral profile is assigned because neither question provides usable behavioral information.

Market Forecasting

Utilizes the Prophet algorithm to forecast the Philippine Stock Exchange Index (PSEi) based on a historical closing-price time series. To accurately model financial asset

behaviors, the prototype employs a multiplicative seasonality model with a changepoint prior scale of 0.05. Yearly seasonality is enabled to capture long-term macroeconomic cycles, while daily and weekly seasonality are disabled to prevent the model from over-fitting on short-term noise. The trained model generates a projection for a 30-business-day horizon, explicitly excluding weekends. Determining the directional trend of the market, the projected market growth is calculated by comparing the last observed PSEi closing price with the final predicted closing price (yhat) at the end of the forecast horizon:

Projected Growth Rate = (Predicted Future Price − Current Price) / Current Price

For integration with the categorical rule matrix, the projected growth forecast is converted into one of four Market Signals. The system evaluates the algorithm's confidence interval to calculate a Forecast Uncertainty Ratio (the width of the interval relative to the predicted price). To assign the final signal, forecast uncertainty is evaluated first in the hierarchy:

1. **Volatile:** the forecast uncertainty ratio exceeds 6%.
2. **Positive:** projected growth is at least +1.0%, provided the Volatile threshold was not triggered.
3. **Negative:** projected growth is at most −1.0%, provided the Volatile threshold was not triggered.
4. **Neutral:** projected growth falls between −1.0% and +1.0%, indicating a flat trend.

The uncertainty ratio is calculated using the width of the Prophet forecast interval

relative to the predicted price on the final forecast date. The Volatile condition is evaluated before directional growth and therefore takes precedence when forecast uncertainty exceeds the established threshold. In this study, the term Volatile refers specifically to high forecast uncertainty, rather than a direct measurement of historical realized market volatility.

Rule-Based Feasibility Assessment

After the Financial Capacity, Behavioral Profile, and Market Signal have been independently generated, their outputs are combined through a complete rule-based feasibility matrix. Financial Capacity contains three categories, Behavioral Profile contains three categories, and Market Signal contains four categories, producing:

3 Behavioral Profiles × 3 Financial Capacities × 4 Market Signals = 36 possible combinations.

Each combination maps to a predefined rule adjustment or Rule Delta. The matrix extends the six initial special-case rules developed during the early design phase into a complete rule base using consistent monotonic decision principles. Higher financial capacity and greater risk tolerance generally permit greater investment exposure, while unfavorable or highly uncertain market conditions reduce feasibility. The six original rules are retained unchanged in the completed matrix. The system begins from a neutral baseline feasibility score of 0.50. Unlike the previous additive design, the final system does not independently add separate market, behavior, and capacity modifiers. Instead, the exact three-variable combination directly selects one rule delta:

$$Feasibility\ Score = 0.50 + \Delta Rule$$

The resulting score is constrained to the range 0.00 to 1.00. The rule simultaneously produces an interpretable System Rule and Rule Result, allowing the final recommendation

to explicitly state how the respondent's behavioral profile, financial capacity, and current market condition contributed to the assessment.

Table 3.2. Positive Market Rule Matrix

Behavioral Profile	Low Financial Capacity	Medium Financial Capacity	High Financial Capacity
Low Risk Tolerance	0.00	+0.05	+0.10
Neutral	+0.05	+0.15	+0.25
High Risk Tolerance	+0.10	+0.25	+0.40

Table 3.3. Neutral Market Rule Matrix

Behavioral Profile	Low Financial Capacity	Medium Financial Capacity	High Financial Capacity
Low Risk Tolerance	-0.15	-0.10	-0.05
Neutral	-0.10	0.0	+0.05
High Risk Tolerance	-0.05	+0.05	+0.15

Table 3.4. Negative Market Rule Matrix

Behavioral Profile	Low Financial Capacity	Medium Financial Capacity	High Financial Capacity
Low Risk Tolerance	-0.40	-0.35	-0.30
Neutral	-0.30	-0.25	-0.20
High Risk Tolerance	-0.20	-0.10	-0.05

Table 3.5. Volatile / High-Uncertainty Market Rule Matrix

Behavioral Profile	Low Financial Capacity	Medium Financial Capacity	High Financial Capacity
Low Risk Tolerance	-0.40	-0.40	-0.40
Neutral	-0.30	-0.20	-0.10
High Risk Tolerance	-0.15	0.00	+0.10

The Rule Delta values in Tables 3.2–3.5 are researcher-defined heuristic decision rules rather than parameters learned from the survey dataset. The matrix was designed to maintain monotonic relationships among financial capacity, behavioral risk tolerance, and market conditions while preserving the initial special-case rules developed during framework design.

Rule Interpretation

Rule deltas are translated into human-readable System Rules and Rule Results. Strong negative adjustments correspond to conservative or defensive guidance, while moderate negative adjustments correspond to cautious preservation. A zero adjustment maintains the baseline and produces balanced guidance. Positive adjustments produce cautious-growth, accumulation, or aggressive-growth guidance depending on the magnitude of the matched rule. This interpretation maintains transparency because the recommendation can be traced directly to one explicit combination in the rule table rather than to an opaque learned decision boundary.

Investment Assessment and Recommendation Module

The recommendation stage does not perform an independent feasibility calculation as the preceding module has already determined the respondent's Feasibility Score, System Rule, and Rule Result using the complete 36-rule matrix; this final stage functions exclusively as an interpretation and presentation layer.

Each determined Rule Result is directly mapped to a corresponding respondent-facing Investment Assessment. These assessments span eight distinct categories based on the magnitude of the rule delta: Conservative Position, Defensive Position, Cautious Preservation, Balanced Position, Cautious Growth, Conservative Growth, Gradual Accumulation, and Growth-Oriented Position. The module subsequently outputs a predefined, explanatory recommendation tailored to the specific assessment, describing the appropriate level of caution, liquidity priority, or investment exposure suitable for the user.

Finally, to support transparency and preserve full traceability between the analytical modules and the final advisory output, the generated recommendation explicitly reports the applied System Rule and the numerical Feasibility Score. This ensures the user is provided with transparent, data-driven reasoning linking their personal financial capacity, behavioral profile, and current market conditions directly to the provided advice.

Validation and Evaluation

Because the proposed framework consists of different analytical methods, evaluation is performed separately for each model using metrics appropriate to its task rather than applying one universal train-test procedure across the entire pipeline.

Prophet Forecasting Evaluation

Prophet forecasting performance is evaluated using time-series cross-validation. The prototype uses an initial historical window of 730 days, a 180-day spacing between cutoff points, and a 30-day forecasting horizon. Forecast accuracy is measured using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). RMSE emphasizes larger forecast errors, MAE reports the average absolute prediction error, and MAPE expresses forecast error relative to the actual values. These metrics are reported as empirical performance measurements rather than being evaluated against an arbitrarily imposed universal accuracy threshold.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - y_{\hat{i}})^2}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - y_{\hat{i}}|$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{A_i - F_i}{A_i}\right|_i \times 100$$

Fuzzy C-Means Evaluation

To validate the Fuzzy C-Means model, it is assessed using the Fuzzy Partition Coefficient (FPC) (ranging from 0 to 1), where a higher score indicates less ambiguous and more concentrated fuzzy memberships [31]. Additionally, the Silhouette Score is utilized to evaluate the geometric separation and compactness of the dominant cluster assignments. Using both metrics allows the study to rigorously evaluate the degree of concentration of the fuzzy memberships alongside the actual structural separation of the resulting financial capacity groups.

XLM-RoBERTa Researcher-Consensus Validation

The open-ended behavioral responses do not contain pre-existing ground-truth risk labels, the XLM-RoBERTa module is evaluated against a human-annotated reference set. A reproducible random sample of 25 respondent profiles is selected using a fixed random seed. The same Q7 and Q9 responses were independently reviewed by three researchers using the behavioral definitions established for the study. Each researcher first assigned an item-level classification to Question 7 and Question 9 as Low Risk Tolerance, Neutral, or High Risk Tolerance. The two item-level classifications were then combined using the same Behavioral Profile Combination and Conflict Resolution Matrix shown in Table 3.1, producing one final human behavioral profile for each researcher.

When at least two researchers assign the same label, the majority category is used as the Researcher Consensus ground truth. Cases without majority consensus are identified for adjudication rather than automatically treated as valid labels. XLM-RoBERTa predictions are compared against the final researcher-consensus labels using accuracy, macro precision, macro recall, macro F1-score, weighted F1-score, a classification report, and a confusion matrix.

$$precision = \frac{TP}{TP+FP}$$

$$recall = \frac{TP}{TP+FN}$$

$$f_1 = 2\times\frac{precision\times recall}{precision+recall}$$

Qualitative Recommendation Evaluation

In addition to model-specific quantitative validation, a qualitative evaluation was

conducted to assess whether the framework's final recommendations were understandable, transparent, relevant, and useful to intended users. This evaluation does not measure predictive accuracy and does not train, modify, or optimize the analytical models. Instead, it evaluates how effectively the final advisory output communicates the reasoning and recommendation produced by the completed framework.

A pilot group of 15 participants reviewed two representative advisory outputs generated by the framework. The number of outputs was intentionally limited to two in order to reduce respondent burden and keep the evaluation practical within the study period. Requiring participants to repeatedly read and rate a larger number of detailed financial advisory scenarios could increase completion time and rating fatigue, potentially reducing the quality of the responses. Consequently, the qualitative evaluation was designed as a focused usability and interpretability assessment rather than a comprehensive evaluation of every possible advisory outcome.

The two advisory outputs were selected from actual combinations produced by the prototype under the prevailing Volatile Market Signal. The first represented a High Financial Capacity respondent with Low Risk Tolerance, producing a Feasibility Score of 0.10 and a Conservative Position. The second represented a High Financial Capacity respondent with Neutral Risk Tolerance, producing a Feasibility Score of 0.40 and a Cautious Preservation assessment. The two examples therefore allowed participants to evaluate recommendations with different behavioral profiles, feasibility scores, and advisory outcomes while keeping the market condition constant.

For each advisory output, participants evaluated four dimensions using a five-point Likert scale, where 1 represented Strongly Disagree and 5 represented Strongly Agree:

1. **Clarity:** the recommendation and its explanation are easy to understand.
2. **Transparency:** the reasoning behind the system's recommendation is understandable.
3. **Relevance:** the recommendation is appropriate to the financial and behavioral profile presented.
4. **Perceived Usefulness:** the recommendation provides useful information for considering a micro-investment decision.

The ratings were summarized using descriptive statistics, including means and standard deviations for each evaluation dimension. Because each participant evaluated both advisory outputs, the evaluation was treated as a pilot assessment of the presentation and interpretability of the recommendation module rather than as evidence of investment profitability or professional financial suitability.

The evaluation was also limited to two representative advisory cases. Therefore, its results should not be generalized to all possible recommendation categories or all 36 combinations in the rule-based feasibility matrix. Feedback obtained from the qualitative evaluation may be used to improve the wording, presentation, and explanation of future advisory outputs. However, the evaluation will not be used to alter the established model classifications or the completed 36-rule feasibility matrix. Furthermore, the results will not be interpreted as evidence of actual investment profitability, professional financial suitability, or long-term investment performance.

Chapter 4

RESULTS AND DISCUSSION

This chapter presents and discusses the results according to the pipeline: financial-capacity classification, behavioral risk classification, market forecasting, the integrated assessment, and model validation. The framework was developed and tested at a controlled, prototype level. All findings describe how the framework behaved on the collected survey sample of 91 responses and the historical PSEi series. However, these numbers are not evidence of investment profitability, professional financial suitability, or long-term investment performance.

The survey underwent several stages of screening and data cleaning. 82 respondents passed the self-reported income range with 69 remaining after the reported income was checked against the scope of the middle-income Filipino group (₱24,060–₱144,360). One respondent lacked the complete core information required for the integrated analytical pipeline and was omitted in the final count of 68 respondents. The validated respondents averaged 43.03 years of age, ₱58,415.26 in monthly gross income, ₱38,028.99 in monthly expenses, and ₱13,175.00 in monthly net savings. Moreover, it should be noted that these figures should not be generalized to all mid-income Filipinos as the collected sample represents a small portion of the whole group.

Table 4.1 Respondent Screening and Final Pipeline Flow

Pipeline Stage	Respondents
Original survey responses	91

Passed self-reported income screening	82
Passed validated numeric income screening	69
Respondents with complete final pipeline information	68

Table 4.1 summarizes the respondent screening process from the original 91 survey responses to the 68 respondents retained for final integration.

Table 4.2 Descriptive Statistics of Income-Validated Respondents

Variable	n	Mean	Standard Deviation	Minimum	Maximum
Age	69	43.03	13.42	20	71
Number of Dependents	69	1.75	1.46	0	6
Monthly Gross Income	69	₱58,415.26	₱33,244.04	₱25,000	₱140,000
Monthly Expenses	69	₱38,028.99	₱25,376.56	₱4,000	₱110,000
Monthly Net Savings	68	₱13,175.00	₱10,091.32	₱0	₱48,000

Table 4.2 provides a descriptive overview of the income-validated sample. Net savings contain 68 observations because one respondent did not provide the complete value required for that variable.

FCM and Financial Capacity Classification

Fuzzy C-Means was applied to the 68 respondents with complete variables (number of dependents, income, expenses, and net savings) using three clusters, a fuzziness

coefficient of 1.5, and a fixed random seed of 42. **Table 4.3** shows that the resulting financial-capacity profiles were reasonably balanced: 21 respondents (30.88%) were assigned to Low, 26 (38.24%) to Medium, and 21 (30.88%) to High Financial Capacity.

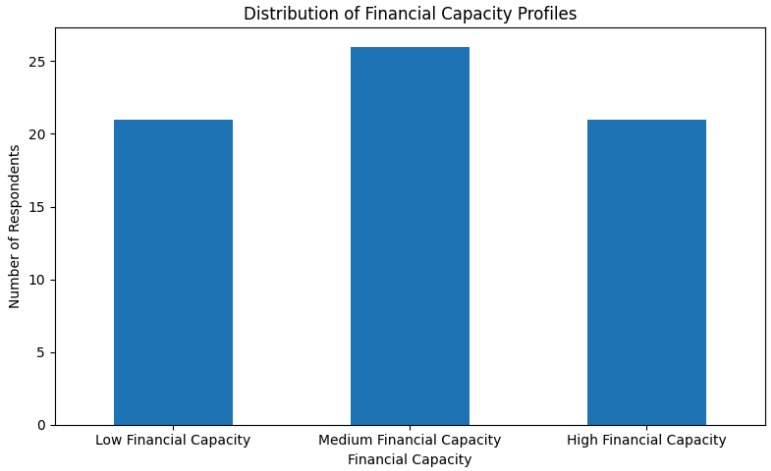


Figure 4.1. Distribution of Financial Capacity Profiles

Table 4.3. Financial Capacity Profile Distribution

Financial Capacity	Respondents	Percentage
Low Financial Capacity	21	30.88%
Medium Financial Capacity	26	38.24%
High Financial Capacity	21	30.88%
Total	68	100%

Table 4.4. Financial Capacity Cluster Characteristics and Interpretation Scores

Financial Capacity	Avg. Dependents	Avg. Monthly Gross Income	Avg. Monthly Expenses	Avg. Net Savings	Expense-to-Income	Savings-to-Income	Capacity Interpretation Score
Low Financial Capacity	2.52	₱33,904.76	₱26,047.62	₱4,400.00	76.83%	12.98%	-2.67
Medium Financial Capacity	2.12	₱91,403.85	₱62,884.62	₱18,096.15	68.80%	19.80%	0.43
High Financial Capacity	0.52	₱42,483.48	₱19,285.71	₱15,857.14	45.40%	37.33%	2.25

As shown in Table 4.4, the Low Financial Capacity cluster exhibited the greatest relative financial strain among the three groups. It had the lowest average monthly gross income, the highest average number of dependents, and the lowest average net savings. Its mean monthly expenses were equivalent to approximately 76.83% of its mean gross income, leaving comparatively limited room for saving or investment. This pattern is consistent with Orbeta's [25] finding that larger household sizes in the Philippines may place greater pressure on household resources and reduce the capacity to save and invest.

In contrast, the Medium Financial Capacity cluster recorded the highest average gross income, but it also had the highest average monthly expenses. The ratio of mean net savings to mean gross income for this cluster was approximately 19.80%. The High Financial Capacity cluster, despite having a lower average income than the Medium cluster, exhibited the strongest overall financial position based on the combined cluster characteristics. It had the lowest average expenses, the fewest dependents, and the highest ratio of mean net savings to mean gross income at approximately 37.33%. Consequently, the High Financial Capacity

cluster obtained the highest Capacity Interpretation Score of 2.25, followed by the Medium cluster at 0.43 and the Low cluster at -2.67.

The expense-to-income and savings-to-income ratios shown in Table 4.4 are descriptive measures calculated from the cluster-level mean values and were not used as additional input variables during Fuzzy C-Means clustering. They are presented only to assist in interpreting the financial characteristics of the resulting clusters.

These labels should therefore be interpreted as relative financial-capacity profiles rather than income brackets. The classification reflects the combined relationship among income, expenses, net savings, and number of dependents rather than gross income alone. An income-only ranking would have placed the Medium Financial Capacity cluster above the High Financial Capacity cluster, illustrating why gross income by itself is insufficient to represent overall financial capacity.

The fuzzy membership values also reveal cases in which a respondent lies near the boundary between clusters. For example, Respondent 2 obtained a membership degree of approximately 0.46 for Low Financial Capacity and 0.49 for High Financial Capacity, indicating substantial overlap between the two profiles. The prototype assigns the respondent to the cluster with the highest membership degree for use in the downstream rule-based feasibility module. However, all membership degrees are retained in the analytical output for reporting and interpretation. At the current stage of the framework, membership uncertainty does not directly modify the Feasibility Score or final recommendation.

XLM-R and Behavioral Classification

The XLM-RoBERTa zero-shot classifier evaluated Question 7 (preference between

safer and higher-risk options) and Question 9 (reaction to a hypothetical loss of ₱3,000 on a ₱10,000 investment) independently. It produced a valid behavioral profile for 68 of the 69 validated respondents, while Respondent 19 lacked sufficient usable information for behavioral classification and was therefore left unclassified. As shown in Figure 4.2 and Table 4.5.1, most respondents were classified as Low Risk Tolerance (44 respondents; 64.71%), followed by Neutral (21; 30.88%) and High Risk Tolerance (3; 4.41%).

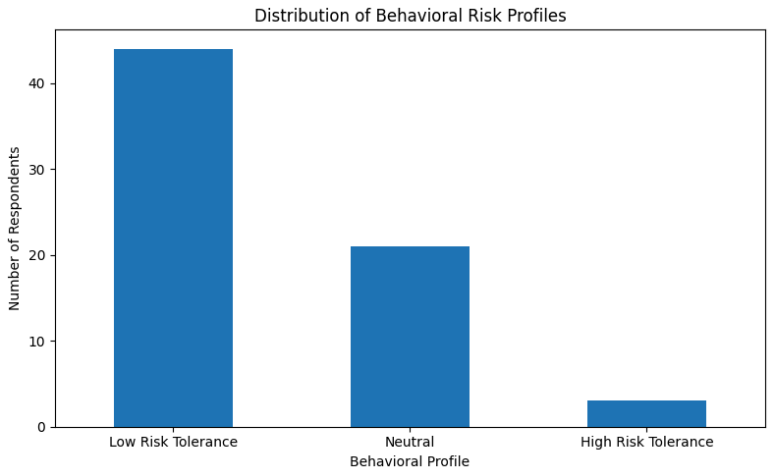


Figure 4.2. Distribution of Final Behavioral Profiles

Table 4.5.1 Quantitative Summary of XLM-RoBERTa Behavioral Outputs

Behavioral Profile	Respondents	Confidence n	Mean Diagnostic Confidence	Median Diagnostic Confidence	Mean Low Score	Mean Neutral Score	Mean High Score
Low Risk Tolerance	44	44	0.6480	0.6206	0.4993	0.3442	0.1566
Neutral	21	17	0.5084	0.4911	0.3066	0.4598	0.2336
High Risk Tolerance	3	3	0.5542	0.4904	0.2252	0.4138	0.3610

Table 4.5.2 Example Numerical XLM-RoBERTa Behavioral Classification

Respondent	Question	Low Score	Neutral Score	High Score	Item Classification	Final Profile
33	Q7	0.9943	0.0055	0.0002	Low Risk Tolerance	Low Risk Tolerance
33	Q9	0.1499	0.7017	0.1484	Neutral	Low Risk Tolerance
45	Q7	0.0977	0.8929	0.0094	Neutral	Neutral
45	Q9	0.2981	0.4250	0.2769	Neutral	Neutral
68	Q7	0.0971	0.0806	0.8223	High Risk Tolerance	High Risk Tolerance
68	Q9	0.1499	0.7017	0.1484	Neutral	High Risk Tolerance

Table 4.5.2 illustrates how the open-ended behavioral responses were numerically processed by XLM-RoBERTa. Each response received a score for the three candidate behavioral profiles, and the highest item-level score determined the classification of Q7 or Q9. The two item classifications were subsequently combined using the Behavioral Profile Combination and Conflict Resolution Matrix rather than by directly averaging their scores.

In addition to the categorical Behavioral Profile, the XLM-RoBERTa module retained numerical zero-shot scores for Low Risk Tolerance, Neutral, and High Risk Tolerance for each classified behavioral response. These scores provide a quantitative representation of the model's semantic classification before the Q7 and Q9 outputs are combined by the conflict-resolution procedure. The highest score for each question determines its item-level classification, while the corresponding winning score is retained as diagnostic confidence.

Consequently, the final Low, Neutral, and High categories reported in Tables 4.5 originate from numerical model outputs rather than from manually assigned numeric encodings.

Two patterns were observed. First, the sample leaned strongly toward low risk, and only three respondents were profiled with High Risk Tolerance. This number is consistent with the low financial literacy rates and limited trust in automated financial advice reported in [10][18]. However, it should be noted that this survey cannot show that these factors caused the pattern. As a result, High Risk Tolerance rows of the rule matrix were rarely exercised in this dataset.

Second, Question 9 rarely separated respondents: 42 of the 66 respondents who provided usable responses to both questions were classified as Neutral on Question 9 because they indicated that they would hold, observe, or wait after a loss. Because the decision rules let a directional answer override a Neutral one, Question 7 effectively decided most of the Low Risk profiles. A total of 31 (70.45%) of the 44 counts came from this precedence rule rather than from agreement between the two items (24 with a Low Question 7 and a Neutral Question 9, and 7 with the reverse). This means that the size of the Low Risk Tolerance group depends substantially on how Question 7 responses are interpreted. Because many responses were short or ambiguous, the behavioral distribution is sensitive to classification errors in these responses. The researcher-consensus validation therefore evaluates how closely the model classifications correspond with independent human interpretation.

Prophet for Market Forecasting

The Prophet model was trained on 1,393 daily PSEi trading-day closing observations from January 4, 2021 to August 28, 2026 and generated a 30-business-day forecast from

August 31 to October 9, 2026. At the end of the horizon, the model predicted a level of 6,025.18 with a projected change of +1.16% in comparison to the last observed close of 5,956.33. The forecast interval at the final date ran from 5,771.44 to 6,285.87 (Prophet's 80% default interval), a width equal to 8.54% of the predicted value.

Table 4.6. PSEi 30-Business-Day Prophet Forecast Summary

Metric	Result
Last Historical Date	August 28, 2026
Last Observed PSEi Close	5,956.33
Forecast End Date	October 9, 2026
Forecasted PSEi Level	6,025.18
Projected Change	+1.16%
Forecast Lower Bound	5,771.44
Forecast Upper Bound	6,285.87
Forecast Uncertainty Ratio	8.54%
Final Market Signal	Volatile

Table 4.6 summarizes the forecast values used to determine the final Market Signal. Although the projected increase of 1.16% satisfies the framework's directional threshold for a Positive signal, the 8.54% forecast uncertainty ratio exceeds the predefined 6% threshold. Because uncertainty is evaluated first, the final Market Signal was classified as Volatile.

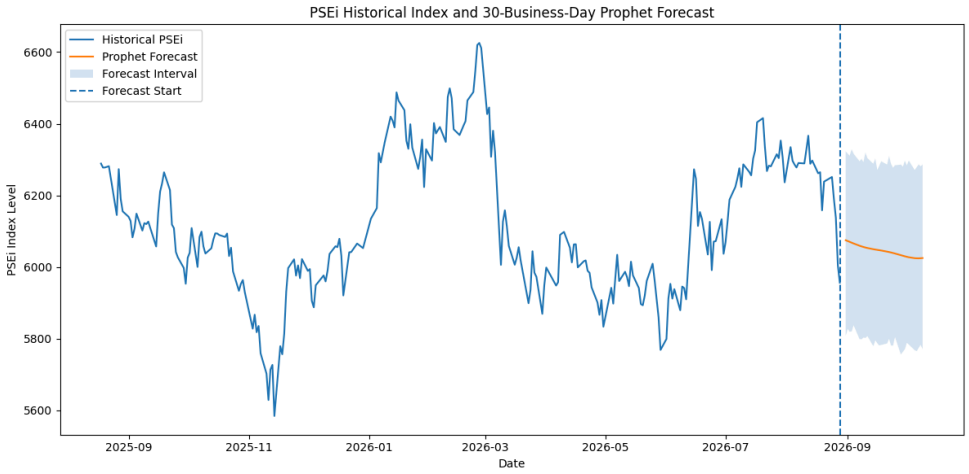


Figure 4.3. Historical PSEi and 30-Business-Day Prophet Forecast

A projected change of +1.16% meets the +1.0% threshold and would ordinarily be classified as Positive. However, uncertainty checks were evaluated first and revealed an 8.54% uncertainty ratio that exceeded the 6% threshold. Because forecast uncertainty is evaluated before directional growth in the framework's classification hierarchy, the resulting Market Signal was classified as Volatile. In this study, Volatile specifically represents high forecast uncertainty and should not be interpreted as a direct measurement of realized historical market volatility.

The projected gain should also be read with caution. The index declined in each of the four sessions before the last observation (about 4.7% from August 24 to August 28). The configured model, however, uses only a trend and yearly seasonality and does not weight this short-term momentum [19][29]. This is why the framework treats the direction of a single point forecast cautiously and lets forecast uncertainty override it.

The 6% forecast-uncertainty threshold is a researcher-defined decision parameter rather than an empirically optimized cutoff. The present study did not evaluate how frequently the Positive, Neutral, Negative, and Volatile signals would be produced across

multiple historical forecast dates. Future work may therefore assess signal frequencies across rolling historical forecasts to determine whether the uncertainty threshold requires further calibration.

Integrated Feasibility and Investment Assessment

The outputs from the three separate modules were combined for the 68 respondents with complete information. Because the Market Signal was constant at Volatile, only the nine rules belonging to the Volatile portion of the 36-rule matrix were eligible for activation. Of these nine rules, eight were actually matched by at least one respondent. The High Risk Tolerance × Low Financial Capacity combination did not occur in the final integrated sample. Therefore, eight of the complete 36 possible rule combinations were exercised during the empirical pipeline run. Table 4.7. Volatile-Market Rule Coverage and Respondent Distribution shows how each eligible Volatile-market combination maps to its Rule Delta, Feasibility Score, Investment Assessment, and observed number of respondents. This stage revealed that 8 of the 9 rules were matched by at least one respondent.

Table 4.7. Volatile-Market Rule Coverage and Respondent Distribution

Behavioral Profile	Financial Capacity	Rule Delta	Score	Assessment	n
Low Risk	Low	-0.40	0.10	Conservative Position	15
Low Risk	Medium	-0.40	0.10	Conservative Position	15
Low Risk	High	-0.40	0.10	Conservative Position	14
Neutral	Low	-0.30	0.20	Defensive Position	6
Neutral	Medium	-0.20	0.30	Defensive Position	9

Neutral	High	-0.10	0.40	Cautious Preservation	6
High Risk	Low	-0.15	0.35	Cautious Preservation	0
High Risk	Medium	0.00	0.50	Balanced Position	2
High Risk	High	+0.10	0.60	Cautious Growth	1

The mean feasibility score was 0.18 (median = 0.10; range = 0.10 to 0.60). A total of 44 respondents (64.71%) received a Conservative Position, 15 (22.06%) received a Defensive Position, 6 (8.82%) Cautious Preservation, 2 (2.94%) a Balanced Position, and 1 (1.47%) Cautious Growth. These were the only five assessment categories that were produced, while none received a Conservative Growth, Gradual Accumulation, or Growth-Oriented Position.

Table 4.8. Final Investment Assessment Distribution

Investment Assessment	Respondents	Percentage
Conservative Position	44	64.71%
Defensive Position	15	22.06%
Cautious Preservation	6	8.82%
Balanced Position	2	2.94%
Cautious Growth	1	1.47%
Total	68	100.0%

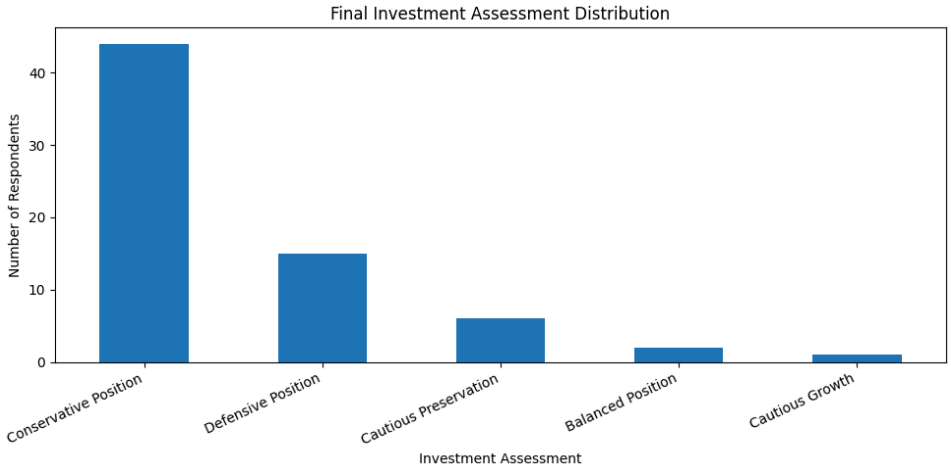


Figure 4.4. Distribution of Final Investment Assessments

These results show that the integration layer works as designed. However, they also expose the limits of personalization under a single market state. The Volatile market signal acted as a constant that pushed everyone toward cautious advice. All 44 Low Risk Tolerance respondents (64.71%) received the same -0.40 adjustment regardless of capacity. And Financial capacity changed the outcome only for the other 24 respondents (35.29%). These findings suggest that the extent of personalization depends heavily on the prevailing market signal.

Table 4.9. Representative Advisory Outputs

Respondent ID	Feasibility Score and System Rule	Investment Assessment	Recommendation
ID 1: High capacity, Low Risk Tolerance, Volatile market	0.10; Strongly decrease	Conservative Position	Prioritize financial stability and capital preservation. Avoid taking substantial investment risk under the current combination of financial capacity, risk tolerance, and market conditions. Maintain sufficient liquid savings and only consider investment exposure when personal finances and market conditions are more favorable.
ID 8: Medium capacity, Neutral, Volatile market	0.30; Decrease	Defensive Position	Maintain a cautious financial position. Focus on liquidity, financial reserves, and lower-risk options rather than significantly increasing investment exposure. Reassess when market conditions or personal

			financial capacity improve.
ID 12: High capacity, Neutral, Volatile market	0.40; Slight decrease	Cautious Preservation	Preserve most available capital while allowing only limited investment exposure if appropriate. Avoid committing funds needed for essential expenses or short-term financial needs.
ID 52: Medium capacity, High Risk Tolerance, Volatile market	0.50; Maintain baseline	Balanced Position	Maintain a balanced approach between financial safety and investment growth. Consider diversified investments using only disposable funds while maintaining adequate savings and liquidity.
ID 68: High capacity, High Risk, Volatile market	0.60; Restricted increase	Cautious Growth	A limited increase in investment exposure may be reasonable, but financial stability should remain the priority. Consider gradual and diversified investing rather than concentrating funds in highly speculative assets.

Each recommendation also traces to one combination of capacity, behavior, and market condition. This responds to transparency being a leading reason why Filipinos hesitate to use robo-advisors [18], in contrast to systems that do not explain their decision processes [2]. However, the Rule Delta values are researcher-defined heuristic parameters rather than values learned from observed financial outcomes. Therefore, the results demonstrate consistent execution of the designed rule system but do not establish that the generated recommendations improve real investment performance or financial outcomes. [5][6].

Model Validation and Evaluation

Each model was evaluated separately using metrics suited to its task.

Prophet was cross-validated by forecasting 30 days ahead from 8 past cutoff dates from February 15, 2023 to July 29, 2026. The module was trained from an initial training window of 730, a spacing of 180 days between cutoffs, and a 30-day horizon. Forecasts deviated from actual values by 331.65 index points on average (MAE), with an RMSE of 366.14 and a MAPE of 5.16%. Given the observed forecast error, small directional changes

near the ±1% classification threshold should be interpreted cautiously. Two caveats apply. First, the validation horizon used in Prophet's cross-validation is 30 calendar days, whereas the deployed forecast covers 30 business days, corresponding to approximately 42 calendar days. Therefore, the reported cross-validation metrics do not perfectly represent the longer calendar duration of the deployed forecast horizon. Second, no baseline forecasting model was evaluated for comparison.

For the Fuzzy C-Means (FCM) module, the algorithm produced a Fuzzy Partition Coefficient (FPC) of 0.7575, which indicates a moderately crisp fuzzy partition (the mathematical minimum for three clusters is 0.33). However, a Silhouette Score of 0.2935 was produced, indicating a relatively weak geometric separation between the dominant cluster assignments. Taken together, the two metrics indicate that the three-cluster solution produced relatively concentrated fuzzy memberships while retaining considerable geometric overlap. This overlap is entirely consistent with the deployment of a fuzzy algorithm, as financial capacity across a sample of 68 mid-income respondents can reasonably be treated as a continuum rather than three strictly separated groups [31]. However, it is noted that the number of clusters was fixed at three (C=3) by design, and alternative algorithms or cluster counts were not comparatively evaluated. A different K value might therefore yield a better-separated structural metric.

Table 4.10. Prophet and Fuzzy C-Means Validation Summary

Module	Metric	Result	Interpretation
Prophet	MAE	331.65	Mean absolute forecast error in PSEi index points
Prophet	RMSE	366.14	Gives greater weight to larger forecast errors
Prophet	MAPE	5.16%	Mean forecast error relative to observed values

Fuzzy C-Means	FPC	0.7575	Relatively concentrated fuzzy memberships
Fuzzy C-Means	Silhouette Score	0.2935	Weak-to-moderate geometric cluster separation

Table 4.10 summarizes the model-specific validation results for Prophet and Fuzzy C-Means. The metrics describe different properties of each analytical module and are therefore interpreted separately rather than combined into a single overall framework accuracy score.

For XLM-RoBERTa, a random sample of 25 respondent profiles were independently evaluated using the behavioral definitions and conflict-resolution procedure established in the study. The resulting researcher-consensus distribution consisted of 13 Low Risk Tolerance, 10 Neutral, and 2 High Risk Tolerance profiles. The XLM-RoBERTa behavioral module matched the final researcher consensus for 21 of the 25 profiles, producing an accuracy of 0.84. The model achieved a Macro Precision of 0.8881, Macro Recall of 0.7487, Macro F1-score of 0.7900, and Weighted F1-score of 0.8362. Per-class F1-scores were 0.85 for Low Risk Tolerance, 0.86 for Neutral, and 0.67 for High Risk Tolerance. The observed accuracy of 0.84 is encouraging given that the model was used in a zero-shot setting without task-specific training; however, the result should be interpreted within the context of the small and imbalanced validation sample [20][21].

Table 4.11. XLM-RoBERTa Researcher-Consensus Validation Metrics

Metric	Result
Evaluated Profiles	25
Correct Classifications	21

Accuracy	0.8400
Macro Precision	0.8881
Macro Recall	0.7487
Macro F1-score	0.7900
Weighted F1-score	0.8362

Table 4.12. XLM-RoBERTa Per-Class Validation Results

Behavioral Profile	Precision	Recall	F1-score	Support
Low Risk Tolerance	0.85	0.85	0.85	13
Neutral	0.82	0.90	0.86	10
High Risk Tolerance	1.00	0.50	0.67	2

The lower performance of the High Risk Tolerance category should be interpreted cautiously because only two High Risk cases were present in the validation sample. Although precision for this category was 1.00, recall was 0.50, meaning that one of the two researcher-labeled High Risk profiles was not identified by the model. Because of this limited class support, the High Risk result is not sufficient to draw strong conclusions regarding the model's ability to identify high-risk investors. The Weighted F1-score of 0.8362 therefore provides a more representative summary of overall classification performance for the evaluated sample, while the Macro F1-score highlights the effect of weaker performance in the underrepresented High Risk category.

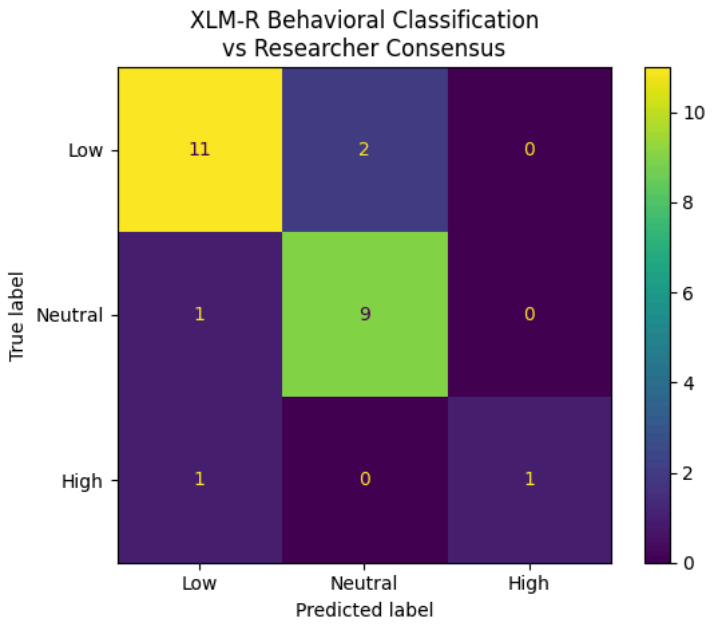


Figure 4.5. Confusion Matrix of XLM-RoBERTa Behavioral Classification Against Researcher Consensus

Four disagreements were observed between the model and the researcher-consensus labels. These occurred for Respondents 1, 9, 24, and 36. Since both the prototype and the human evaluation applied the same deterministic behavioral conflict-resolution logic after Q7 and Q9 had been interpreted, these disagreements indicate differences in the semantic classification of the individual survey responses rather than failures of the conflict-resolution procedure itself. For example, Respondent 1 answered “both half half” for Question 7 and was interpreted by the researchers as Neutral, while the model classified the final profile as Low Risk Tolerance. Respondent 36 stated a preference for investing in stocks while indicating that they would “observe and wait” after a loss. Under the established resolution rules, a High Risk classification for Question 7 combined with a Neutral classification for Question 9 would result in a final High Risk Tolerance profile; however, the model

ultimately classified this respondent as Low Risk Tolerance. This suggests that the disagreement occurred during the zero-shot semantic interpretation of one or both responses before the deterministic combination stage.

Table 4.13. XLM-RoBERTa and Researcher-Consensus Disagreements

Respondent ID	Q7 Risk Preference	Q9 Loss Reaction	Researcher Consensus	Model Classification
1	"both half half"	"no need for a panic… I can wait…"	Neutral	Low Risk Tolerance
9	"I prefer and lean more towards traditional options…"	"I would observe and wait"	Low Risk Tolerance	Neutral
24	"put my money in the bank"	"it happens… look for other investments"	Low Risk Tolerance	Neutral
36	"I'd rather invest in stock because banks has…"	"observe and wait"	High Risk Tolerance	Low Risk Tolerance

The disagreement cases indicate that most classification errors were attributable to the semantic interpretation of short or ambiguous responses rather than to the deterministic Q7–Q9 conflict-resolution procedure.

Three limitations should therefore be considered when interpreting the behavioral validation results. First, the validation sample contained only 25 profiles and included only two High Risk Tolerance cases, resulting in substantial class imbalance. Second, the researchers who developed the behavioral coding definitions also participated in the annotation process, which may introduce methodological bias toward the study's predefined

categories. Third, no formal inter-rater reliability statistic was computed among the three researchers. Future validation may use a larger and more balanced independently annotated dataset and may calculate an inter-rater agreement measure before establishing the final consensus labels.

Qualitative Recommendation Evaluation

A pilot qualitative evaluation was conducted to assess the clarity, transparency, relevance, and perceived usefulness of the framework's final advisory output. Fifteen participants each reviewed two representative recommendations generated by the completed framework. Both advisory examples were based on actual combinations produced under the prevailing Volatile Market Signal. The first represented a High Financial Capacity profile with Low Risk Tolerance, resulting in a Feasibility Score of 0.10 and a Conservative Position. The second represented a High Financial Capacity profile with Neutral Risk Tolerance, resulting in a Feasibility Score of 0.40 and a Cautious Preservation assessment. These two cases were selected to expose participants to different behavioral profiles, feasibility scores, and recommendation outcomes while keeping the market condition constant.

Each participant rated both advisory outputs using a five-point Likert scale, where 1 represented Strongly Disagree and 5 represented Strongly Agree. Four dimensions were evaluated: clarity, transparency, relevance, and perceived usefulness. Because each participant evaluated two advisory outputs, the two ratings provided by each participant for the same dimension were first averaged. The descriptive statistics shown in **Table 4.14** were then calculated across the resulting 15 participant-level averages. This approach retains the participant, rather than the individual advisory rating, as the unit of analysis.

Table 4.14. Qualitative Recommendation Evaluation Summary

Dimensions	N	Mean	Median	SD
Clarity	15	4.57	5.0	0.50
Transparency	15	4.57	4.5	0.42
Relevance	15	4.43	4.5	0.46
Perceived Usefulness	15	4.57	4.5	0.46

The results indicate a generally positive evaluation of the advisory outputs. Clarity, transparency, and perceived usefulness each obtained a mean rating of 4.57, while relevance obtained a mean of 4.43. All individual ratings were either 4 or 5, indicating that participants generally found the recommendations understandable, transparent, relevant, and useful. Relevance received the lowest mean, although it remained favorable. One participant suggested providing examples of suitable investments and possible investment amounts to make the advice more actionable; however, such specific recommendations fall outside the scope of the current prototype.

These findings should be interpreted as a pilot evaluation because only 15 participants assessed two advisory outputs under the same Volatile Market Signal. Therefore, the results should not be generalized to all possible outputs of the 36-rule matrix or interpreted as evidence of investment profitability.

Chapter 5

CONCLUSION

This study created and evaluated a hybrid AI-based financial advisory framework for digital micro-investment decision support among mid-income Filipinos. The framework integrated three analytical components: Fuzzy C-Means for relative Financial Capacity profiling, zero-shot XLM-RoBERTa for Behavioral Risk classification, and Prophet for PSEi market forecasting. The resulting Financial Capacity, Behavioral Profile, and Market Signal were integrated through a complete 36-rule feasibility matrix that generated an interpretable Feasibility Score, Investment Assessment, and user-facing recommendation.

The prototype ran end to end on the collected sample, with every recommendation traceable to its corresponding Financial Capacity, Behavioral Profile, Market Signal, matched rule, and Feasibility Score. This traceability supports greater transparency in the decision-making and recommendation process compared with an opaque end-to-end model. The framework successfully combined financial, behavioral, and market information; however, the Market Signal was classified as Volatile for the entire final cohort. This influenced the system toward more cautious recommendations and limited the range of personalization observed during the experimental run. The individual analytical modules were also evaluated separately. Fuzzy C-Means produced an FPC of 0.7575 and a Silhouette Score of 0.2935, indicating relatively concentrated fuzzy memberships despite overlap among the capacity groups. The XLM-RoBERTa behavioral module transformed the open-ended Q7 and Q9 responses into numerical Low, Neutral, and High zero-shot classification scores before applying the transparent behavioral conflict-resolution procedure. Its resulting behavioral classifications agreed with the researcher consensus on 21 of 25

validation profiles, producing an accuracy of 0.84, a Macro F1-score of 0.79, and a Weighted F1-score of 0.8362. Prophet produced a MAPE of 5.16%, while its final forecast had an uncertainty ratio of 8.54%, resulting in the Volatile Market Signal. The qualitative pilot evaluation further showed favorable user perceptions of the advisory outputs, with mean ratings of 4.57 for Clarity, 4.57 for Transparency, 4.43 for Relevance, and 4.57 for Perceived Usefulness.

These findings support the framework as a prototype-level implementation of an interpretable financial decision-support system. The sample and validation sets were relatively small, the behavioral validation contained only a limited number of High Risk Tolerance cases, and the Rule Delta adjustments and market-signal thresholds were researcher-defined rather than learned from financial outcomes. In addition, Prophet was not compared against a forecasting baseline, and the qualitative evaluation involved only two representative advisory outputs under the same volatile market condition. Therefore, although the framework behaved consistently and transparently according to its defined rules, the study does not establish that its recommendations are optimal, professionally suitable, or capable of improving actual financial or investment outcomes. The framework should instead be interpreted as a research prototype for generating traceable micro-investment decision-support guidance.

Future work could incorporate fuzzy membership degrees directly into the feasibility matrix so that borderline respondents are treated more gradually rather than through dominant-cluster assignment alone. The researcher-defined Volatile threshold and Rule Delta values could also be calibrated using additional historical data and repeated market simulations. Prophet may be compared against a simple forecasting baseline and evaluated

using a validation horizon that more closely matches the deployed 30-business-day forecast. Repeating the behavioral validation with a larger, more balanced, and independently annotated sample, together with a formal inter-rater reliability measure, would further strengthen the evaluation. Future qualitative testing may also include more participants and a wider variety of advisory outputs representing different market signals and recommendation categories.

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APPENDIX A

Complete 36-Rule Feasibility Matrix

Table A.1 Complete 36-Rule Feasibility and Investment Assessment Matrix

#	Market Signal	Behavioral Profile	Financial Capacity	Rule Delta	Feasibility Score	System Rule	Investment Assessment
1	Negative	Low Risk	Low	-0.40	0.10	Strongly decrease	Conservative Position
2	Negative	Low Risk	Medium	-0.35	0.15	Strongly decrease	Conservative Position
3	Negative	Low Risk	High	-0.30	0.20	Decrease	Defensive Position
4	Negative	Neutral	Low	-0.30	0.20	Decrease	Defensive Position
5	Negative	Neutral	Medium	-0.25	0.25	Decrease	Defensive Position
6	Negative	Neutral	High	-0.20	0.30	Decrease	Defensive Position
7	Negative	High Risk	Low	-0.20	0.30	Decrease	Defensive Position
8	Negative	High Risk	Medium	-0.10	0.40	Slight decrease	Cautious Preservation
9	Negative	High Risk	High	-0.05	0.45	Slight decrease	Cautious Preservation
10	Neutral	Low Risk	Low	-0.15	0.35	Slight decrease	Cautious Preservation
11	Neutral	Low Risk	Medium	-0.10	0.40	Slight decrease	Cautious Preservation
12	Neutral	Low Risk	High	-0.05	0.45	Slight decrease	Cautious Preservation
13	Neutral	Neutral	Low	-0.10	0.40	Slight decrease	Cautious Preservation
14	Neutral	Neutral	Medium	+0.00	0.50	Maintain baseline	Balanced Position
15	Neutral	Neutral	High	+0.05	0.55	Restricted increase	Cautious Growth
16	Neutral	High Risk	Low	-0.05	0.45	Slight decrease	Cautious Preservation
17	Neutral	High Risk	Medium	+0.05	0.55	Restricted increase	Cautious Growth
18	Neutral	High Risk	High	+0.15	0.65	Slight increase	Gradual Accumulation
19	Positive	Low Risk	Low	+0.00	0.50	Maintain baseline	Balanced Position
20	Positive	Low Risk	Medium	+0.05	0.55	Restricted increase	Cautious Growth
21	Positive	Low Risk	High	+0.10	0.60	Slight increase	Conservative Growth
22	Positive	Neutral	Low	+0.05	0.55	Restricted increase	Cautious Growth

#	Market Signal	Behavioral Profile	Financial Capacity	Rule Delta	Feasibility Score	System Rule	Investment Assessment
23	Positive	Neutral	Medium	+0.15	0.65	Slight increase	Gradual Accumulation
24	Positive	Neutral	High	+0.25	0.75	Slight increase	Gradual Accumulation
25	Positive	High Risk	Low	+0.10	0.60	Restricted increase	Cautious Growth
26	Positive	High Risk	Medium	+0.25	0.75	Slight increase	Gradual Accumulation
27	Positive	High Risk	High	+0.40	0.90	Strongly increase	Growth-Oriented Position
28	Volatile	Low Risk	Low	-0.40	0.10	Strongly decrease	Conservative Position
29	Volatile	Low Risk	Medium	-0.40	0.10	Strongly decrease	Conservative Position
30	Volatile	Low Risk	High	-0.40	0.10	Strongly decrease	Conservative Position
31	Volatile	Neutral	Low	-0.30	0.20	Decrease	Defensive Position
32	Volatile	Neutral	Medium	-0.20	0.30	Decrease	Defensive Position
33	Volatile	Neutral	High	-0.10	0.40	Slight decrease	Cautious Preservation
34	Volatile	High Risk	Low	-0.15	0.35	Slight decrease	Cautious Preservation
35	Volatile	High Risk	Medium	+0.00	0.50	Maintain baseline	Balanced Position
36	Volatile	High Risk	High	+0.10	0.60	Restricted increase	Cautious Growth

Note. Feasibility Score = 0.50 + Rule Delta, constrained to the range 0.00 to 1.00. Financial Capacity is Low, Medium, or High; Behavioral Profile is Low Risk, Neutral, or High Risk Tolerance. Rule deltas are researcher-defined heuristics rather than learned parameters.

Table A.2 Rule Result, Investment Assessment, and Recommendation Templates

Rule Result	Investment Assessment	Recommendation Advice
Conservative advice	Conservative Position	Prioritize financial stability and capital preservation. Avoid taking substantial investment risk under the current combination of financial capacity, risk tolerance, and market conditions. Maintain sufficient liquid savings and only consider investment exposure when personal finances and market conditions are more favorable.
Defensive advice	Defensive Position	Maintain a cautious financial position. Focus on liquidity, financial reserves, and lower-risk options rather than significantly increasing investment exposure. Reassess when market conditions or personal financial capacity improve.
Cautious preservation advice	Cautious Preservation	Preserve most available capital while allowing only limited investment exposure if appropriate. Avoid committing funds needed for essential expenses or short-term financial needs.
Balanced advice	Balanced Position	Maintain a balanced approach between financial safety and investment growth. Consider diversified investments using only disposable funds while maintaining adequate savings and liquidity.
Cautious advice	Cautious Growth	The current conditions allow some investment exposure, but financial capacity places an important limit on how aggressively funds should be committed. Prefer gradual, diversified investment using only disposable funds.
Cautious growth advice	Cautious Growth	A limited increase in investment exposure may be reasonable, but financial stability should remain the priority. Consider gradual and diversified investing rather than concentrating funds in highly speculative assets.
Conservative growth advice	Conservative Growth	Financial capacity may support limited growth-oriented investment, but the respondent's low risk tolerance should remain the primary constraint. Favor diversified and lower-volatility investments rather than aggressive exposure.

Accumulation advice	Gradual Accumulation	The respondent's profile supports gradually increasing investment exposure. Consider diversified investments and incremental allocation while retaining sufficient liquid funds for financial obligations and emergencies.
Aggressive advice	Growth-Oriented Position	The combined financial, behavioral, and market profile supports a more growth-oriented investment approach. Higher-risk investments may be considered using disposable investment funds, preferably through diversification rather than concentrated exposure.

APPENDIX B

Financial Advisory Prototype Framework Implementation

The Python notebook used to run the Fuzzy C-Means, XLM-RoBERTa, Prophet, and rule-based modules is available at the link below.

Link:

APPENDIX C

Data Collection Survey Questionnaires Format

Table C.1 Main Data Collection Survey Questionnaire Format (May 2026 - Sep 2026)

Section	Item	Question	Response format
Screening and consent	1	Do you have a monthly gross income range between ₱24,060–₱144,360?	Yes / No (No ends the survey)
	2	I have read and understood the nature, purpose, and data privacy notice of this research. I confirm that my participation is voluntary and that I agree to proceed with this survey.	Yes / No (No ends the survey)
Demographics	3	Age	Short answer
	4	Sex	Male / Female

	5	Civil Status	Single / Married / Separated / Widowed
	6	Employment Status	Employed (Full-time) / Employed (Part-time) / Self-employed/Freelancer / Business Owner / Retired / Other
	7	Educational Attainment	Elementary Graduate / Junior High School Graduate / Senior High School Graduate / Bachelor's Degree / Master's Degree / Doctoral Degree (PhD) / Other
	8	Number of Dependents	Short answer
Finances	9	1. Average Monthly Gross Income	Short answer (₱)
	10	2. Average Monthly Expenses	Short answer (₱)
	11	3. Average Monthly Net Savings	Short answer (₱)
	12	4. Total Average Educational Expenses (for further studies or of dependents)	Short answer (₱ or N/A)
	13	5. Do you have current loans/debts? (If yes, please specify the estimated total amount)	Short answer (₱ or N/A)
Behavioral	14	1. How often do you set aside money for savings?	Always / Often / Sometimes / Rarely / Never
	15	2. How often do you actively look for investment opportunities?	Always / Often / Sometimes / Rarely / Never
	16	3. How often do you still have money left at the end of the month?	Always / Often / Sometimes / Rarely / Never
	17	4. Do you spend money based on plans or impulse?	Mostly planned / Both planned and impulsive / Mostly impulsive / Other

	18	5. What would you do if you had extra money?	Save it / Spend it / Invest it / Split between saving and spending / Other
	19	6. When making financial decisions, which best describes you?	I rely on instinct / I seek advise from friends or family / I research before deciding / I follow a structured financial plan / Other
	20	7. How do you feel about investing in high-risk, high-reward platforms (like Crypto or Stocks) compared to keeping your money in a traditional bank? (Do you strictly prefer one over the other, or do you prefer to put half in the bank and half in investments?)	Open-ended
	21	8. What is your primary goal for investing your money, and how fast do you want to see returns? (If you don't have a specific goal yet, or if you prefer a mix of fast and slow returns, you can state that as well.)	Open-ended
	22	9. Imagine you invested ₱10,000, and after one week, its value dropped to ₱7,000. What would you do and how would you feel? (e.g., Would you panic and sell everything, buy more because it's cheap, or just observe and wait?)	Open-ended

Table C.2 Pilot Survey Questionnaire Format

Section	Item	Question	Response format
Sample Advisory Output #1	1	Clarity: The recommendation and its explanation are easy to understand.	1 — Strongly Disagree / 2 — Disagree / 3 — Neutral / 4 — Agree / 5 — Strongly Agree
	2	Transparency: The reasoning behind the system's recommendation is understandable.	1 — Strongly Disagree / 2 — Disagree / 3 — Neutral / 4 — Agree / 5 — Strongly Agree

	3	Relevance: The recommendation is appropriate to the financial and behavioral profile presented.	1 — Strongly Disagree / 2 — Disagree / 3 — Neutral / 4 — Agree / 5 — Strongly Agree
	4	Perceived Usefulness: The recommendation provides useful information for considering a micro-investment decision.	1 — Strongly Disagree / 2 — Disagree / 3 — Neutral / 4 — Agree / 5 — Strongly Agree
	5	Optional: What would make this recommendation clearer or more useful?	Open-ended
Sample Advisory Output #2	6	Clarity: The recommendation and its explanation are easy to understand.	1 — Strongly Disagree / 2 — Disagree / 3 — Neutral / 4 — Agree / 5 — Strongly Agree
	7	Transparency: The reasoning behind the system's recommendation is understandable.	1 — Strongly Disagree / 2 — Disagree / 3 — Neutral / 4 — Agree / 5 — Strongly Agree
	8	Relevance: The recommendation is appropriate to the financial and behavioral profile presented.	1 — Strongly Disagree / 2 — Disagree / 3 — Neutral / 4 — Agree / 5 — Strongly Agree
	9	Perceived Usefulness: The recommendation provides useful information for considering a micro-investment decision.	1 — Strongly Disagree / 2 — Disagree / 3 — Neutral / 4 — Agree / 5 — Strongly Agree
	10	Optional: What would make this recommendation clearer or more useful?	Open-ended

APPENDIX D

Main Survey Data Collection Dataset (May 2026 - Sep 2026)

The survey dataset used in this study (91 responses collected from May to September 2026) is available at the link below.

Link:

APPENDIX E

Pilot Survey Data Collection Dataset (May 2026 - Sep 2026)

Table E.1 Pilot Survey Data Collection

Section	Respondent	Clarity	Transparency	Relevance	Usefulness
Sample Advisory Output #1	1	4	5	5	4
	2	4	4	4	4
	3	4	4	4	4
	4	4	4	4	4
	5	5	5	4	4
	6	5	4	4	5
	7	4	4	4	4
	8	5	5	5	5
	9	5	5	5	5
	10	4	5	4	4
	11	5	4	5	5
	12	4	4	5	5
	13	5	5	5	5
	14	5	5	4	4
	15	5	5	5	5
Sample Advisory Output #2	1	4	5	4	5
	2	4	4	4	4
	3	4	4	4	4
	4	4	4	4	4
	5	5	5	5	5
	6	5	5	4	5
	7	4	4	4	4

	8	5	5	5	5
	9	5	5	5	5
	10	4	5	4	4
	11	5	5	5	5
	12	5	5	4	5
	13	5	5	5	5
	14	5	4	4	5
	15	5	4	5	5

Table E.2 Pilot Survey Data Collection Optional Feedback

Respondent	Sample Output	Optional Comment
8	#1	Good
11	#1	This recommendation is clear and useful to me because it is based on people's actual financial situation. The recommendations are also practical and helpful, making them easier for us to understand and apply.
11	#2	It would be more useful if the recommendation included specific examples of suitable investment options and a clear explanation of how much money could reasonably be invested.